

Multiple Choice

1. **Answer:** C

Options: A) New matter was created in the bubbles.; B) Old matter was destroyed as the water turned to gas.; C) Matter was conserved when it changed form.; D) The composition of the matter changed.

Note: The correct answer is based on the principle of conservation of mass, which states that matter cannot be created or destroyed in a physical change, such as the boiling of water, where it transforms from liquid to gas while the total amount of matter remains constant.

2. **Answer:** A

Options: A) Traits from two parents are combined.; B) More organisms reproduce this way.; C) Offspring grow in different environments.; D) Offspring come from identical parents.

Note: The correct answer is based on the fact that sexual reproduction involves the combination of genetic material from two parents, leading to offspring with a unique mix of traits, whereas asexual reproduction produces genetically identical offspring.

3. **Answer:** A

Options: A) Volvox colonies produce gametes.; B) Volvox cells undergo conjugation.; C) Volvox colonies produce single-celled offspring.; D) Volvox cells reproduce by vegetative budding.

Note: The correct answer is based on the fact that volvox, as a colonial organism, forms specialized reproductive cells in the form of gametes within its colonies, whereas paramecium typically reproduces sexually through a process of conjugation without forming gametes.

4. **Answer:** B

Options: A) increases the supply of resources; B) decreases the amount of resources; C) results in less demand for resources; D) improves the quality of the resources

Note: The increase in the human population leads to greater demand for natural resources such as water, food, and energy, resulting in a depletion of these essential resources.

5. **Answer:** C

Options: A) the phases of the Moon.; B) the tides on Earth.; C) seasons on Earth.; D) lunar eclipses.

Note: The seasons on Earth are primarily caused by the tilt of Earth's axis relative to its orbit around the Sun, rather than being directly influenced by the interactions between Earth and the Moon.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Talons in hawks are not acquired characteristics; they are inherited traits that have evolved over generations through natural selection to aid in hunting.

7. **Answer:** True

Options: A) True; B) False

Note: Notifying the teacher ensures that the spill is handled safely and appropriately, preventing potential hazards and ensuring proper cleanup procedures are followed.

8. **Answer:** True

Options: A) True; B) False

Note: The statement is true because multicellular organisms that are autotrophic can be classified in both the kingdom of Protists, which includes some multicellular algae, and the kingdom of Plants, which consists of all multicellular autotrophic organisms.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because balloons float upward due to buoyancy, which is the upward force exerted by a fluid, rather than being influenced by gravity, which pulls objects downward.

10. **Answer:** True

Options: A) True; B) False

Note: This statement is true because verifying the accuracy of the data and ensuring that the experiment is repeatable are essential steps in the scientific method to validate findings and eliminate potential errors.

Long Form Response

11. **Answer:** When a plastic rod is rubbed with fur, a process called triboelectric charging occurs. During this interaction, electrons, which are negatively charged particles, are transferred from the fur to the plastic rod. As a result, the rod gains extra electrons, leading to an overall negative charge. The fur, having lost some electrons, becomes positively charged. This movement of electrons is what generates static electricity and is a key concept in understanding how materials can become charged through friction.
Note: correct because it accurately describes the triboelectric charging process, where rubbing the plastic rod with fur causes the transfer of electrons, resulting in the rod becoming negatively charged and the fur becoming positively charged.
12. **Answer:** The communication process within cells, where molecules relay messages from the cell membrane to the nucleus, resembles the functioning of the nervous system. Just as neurons transmit signals to communicate information throughout the

body, cellular signaling pathways enable cells to respond to external stimuli and coordinate activities. For instance, when a hormone binds to a receptor on the cell membrane, it triggers a cascade of molecular events that ultimately influence gene expression in the nucleus. This parallel highlights how both systems rely on precise communication to maintain homeostasis and respond effectively to changes in their environment. Understanding this connection illustrates the importance of signaling in both cellular and bodily functions.

Note: correct because it effectively illustrates the parallel between cellular communication, where signals are relayed to influence gene expression, and the nervous system's function of transmitting signals to coordinate bodily responses.

End of Answer Key